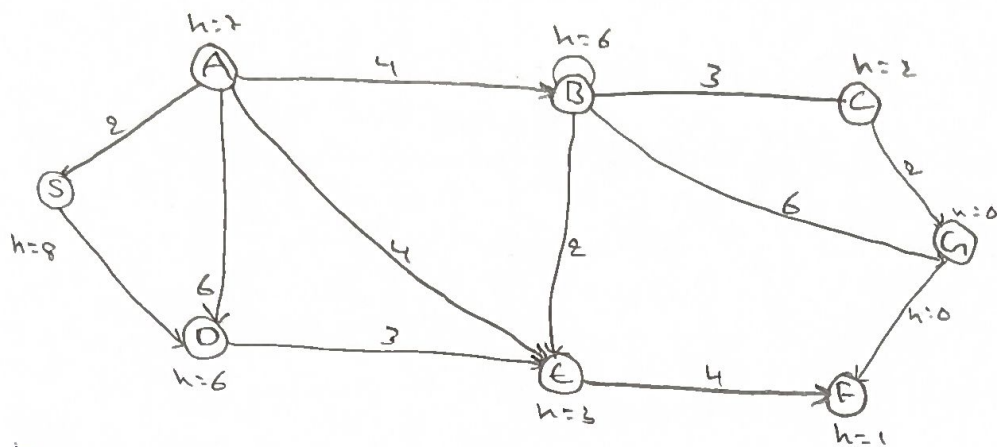


Smart Ambulance Navigation



By Using formula is

$$f(x) = g(n) + h(n)$$

Start node is $\rightarrow S$ and Goal is $\rightarrow C$

Start

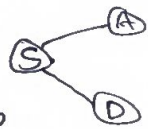
1. S

$$g(S) = 0$$

$$h(S) = 8$$

$$f(S) = 0 + 8 = 8$$

2. A



$$g(A) = 2$$

$$h(A) = 7$$

$$f(A) = 2 + 7 = 9$$

D

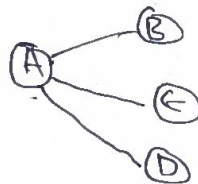
$$g(D) = 6$$

$$h(D) = 6$$

$$f(D) = 6 + 6 = 12$$

$f(A)$ is smallest is 9

3. A



$$g(B) = 4$$

$$h(B) = 6$$

$$f(B) = 4 + 6 = 10$$

E

$$g(E) = 4$$

$$h(E) = 3$$

$$f(E) = 4 + 3 = 7$$

D

$$g(D) = 6$$

$$h(D) = 6$$

$$f(D) = 6 + 6 = 12$$

$f(E)$ is a smallest is 7

A path is



20

100

10/8/20

8

4. Step

Expand $E \xrightarrow{B} F$

B

$$g(B) = 2$$

$$h(B) = 6$$

$$f(B) = 6 + 2 = 8$$

F

$$g(F) = 4$$

$$h(F) = 1$$

$$f(F) = 4 + 1 = 5$$

$\therefore f(F)$ is a smallest path

6. Step

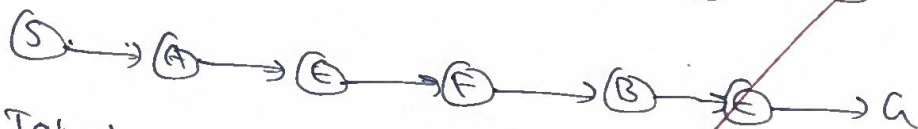
Expand C

$$g(C) = 9 + 2 = 11$$

$$h(C) = 0$$

$$\therefore f(C) = 11$$

$\therefore$ selected path from $S \rightarrow C$ is



Total minimum cost = 11

Best route (minimum path is)



5. step

Expand $B \xrightarrow{C} G$

$$g(C) = 6 + 3 = 9$$

$$h(C) = 2$$

$$f(C) = 9 + 2 = 11$$

G

$$g(G) = 6 + 6 = 12$$

$$h(G) = 0$$

$$f(G) = 12$$

Autonomous warehouse Robot :-

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

The given data

Start : (5, 1) Goal (1, 7)

by using

$$h(u) = |r - r_g| + |c - c_g|$$

1. Possible

(5, 1) → (5, 2) → (5, 3) → (5, 4) → (5, 5) → (5, 6) → (5, 7).

⇒ (5, 7) → (4, 7) and (3, 7) is an obstacle.

2. Replanning the path is

(4, 7) → (4, 6) → (3, 6) (obstacle).

Again updating is

(3, 6) → (3, 5) → (3, 4)

then

(3, 4) → (2, 4) → (1, 4)

Finally it's

(1, 4) → (1, 5) → (1, 6) → (1, 7)

∴ Final path is without any obstacle.
is.

(1, 4) → (1, 5) → (1, 6) → G

3. University Exam Scheduling

a. CSP Formulation is.

Given,

$$X = \{E_1, E_2, E_3, E_4, E_5, E_6\}$$

Each must be assigned a time slot.

it Domains.

Initially

$$D(E_1) = D(E_2) = \dots = D(E_6) = \{T_1, T_2, T_3, T_4\}$$

$\therefore$ Half Capacities from data

$$A = 120, \quad B = 90, \quad C = 60$$

B. Constraints

$$E_1 \neq E_2$$

$$E_1 \neq E_3$$

$$E_2 \neq E_4$$

$$E_3 \neq E_5$$

$$E_4 \neq E_6$$

$$E_2 \neq E_5$$

$\therefore$ Exam Remaining domain

$$E_1 \longrightarrow \{T_2, T_3, T_4\}$$

$$E_2 \longrightarrow \{T_1, T_2, T_3, T_4\}$$

$$E_4 \longrightarrow \{T_2, T_3, T_4\}$$

$$E_5 \longrightarrow \{T_2, T_3, T_4\}$$

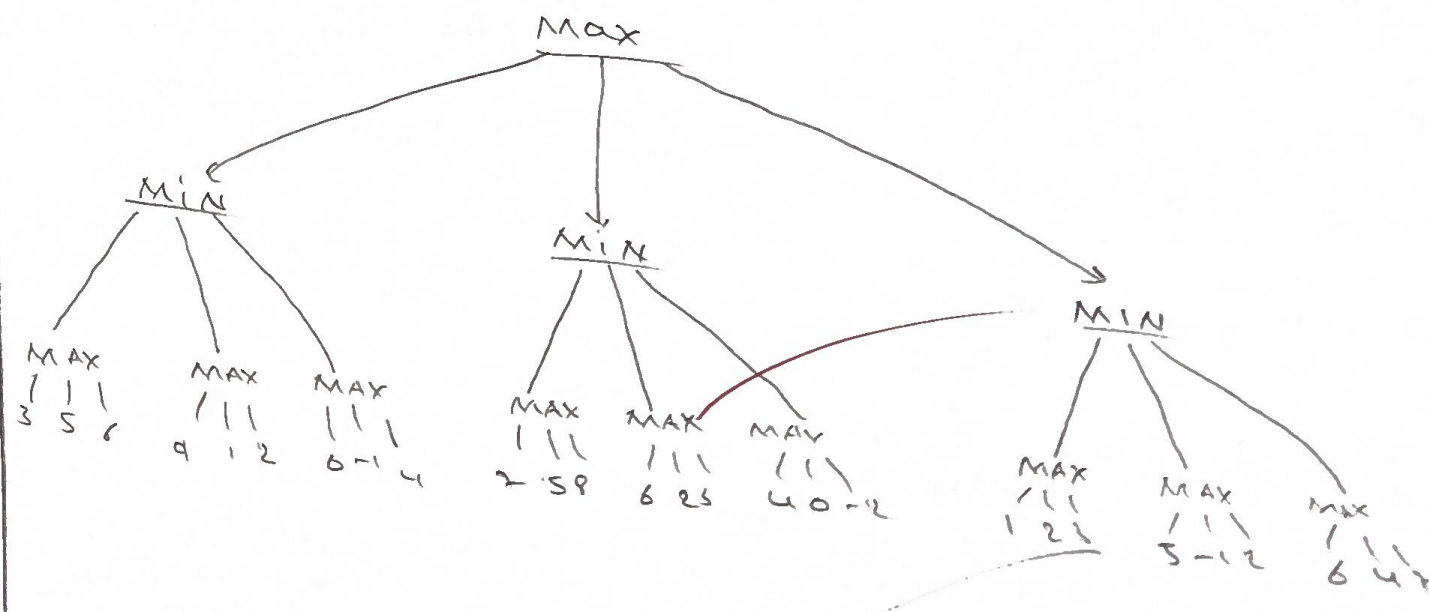
$$E_6 \longrightarrow \{T_1, T_2, T_3, T_4\}$$

Final Timetable is

Time slot	Examination	Hall	Students
T_1	E_2	B	80
T_1	E_3	A	20
T_1	E_6	C	50
T_2	E_1	A	100
T_2	E_4	C	60
T_3	E_5	A	40
T_4			
Final		A	90

$E_1 = T_2, E_2 = T_1, E_3 = T_1, E_4 = T_2, E_5 = T_3, E_6 = T_1.$

4. α - β Pruning



1. Minimax Calculation

Left

$$\max(3, 5, 6) = 6$$

$$\max(9, 1, 2) = 9$$

$$\max(8, -1, 4) = 8$$

$$\min(6, 9, 4) = 4$$

Middle

$$\max(7, 5, 8) = 8$$

$$\max(6, 2, 3) = 6$$

$$\max(4, 0, -2) = 4$$

$$\max(8, 6, 4) = 8$$

Right

$$\max(1, 2, 3) = 3$$

$$\max(5, -1, 2) = 5$$

$$\max(6, 4, 7) = 7$$

$$\max(3, 3, 7) = 7$$

$$\therefore \max \text{ of } (4, 4, 3) = 4$$

Optimum minimax value = 4

2. α - β

Pruning (Left to right)

Left

$$\max(3, 5, 6) = 6 \rightarrow \beta = 6$$

$$\max(9, 1, 2) = 9, \quad \alpha = 9$$

$$\alpha(9) \geq \beta(6) \rightarrow \text{Prune}$$

$$\max(8, -1, 4) = 8$$

$$\therefore \min = 4$$

Middle

$$\max(7, 5, 8) = 8$$

$$(6, 2, 3) \rightarrow 6$$

$$(4, 0, -2) \rightarrow 4$$

No pruning

$$\therefore \min \text{ value} = 4$$

Right

$$\max(1, 2, 3) = 3 \rightarrow \beta = 3$$

$$\min(3(3) \leq \alpha(4) \rightarrow \text{Pruning}$$

$$\min = 3$$

$$\therefore \text{Optimum minimax} = 4$$

Delivery route optimization

Distance Matrix (km)

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

neighbouring routes

$$S_1: D \rightarrow A \rightarrow C \rightarrow B \rightarrow E \rightarrow D = 26 \text{ km}$$

$$S_2: D \rightarrow B \rightarrow A \rightarrow C \rightarrow E \rightarrow D = 30 \text{ km}$$

$$S_3: D \rightarrow A \rightarrow B \rightarrow E \rightarrow C \rightarrow D = 24 \text{ km}$$

$$S_4: D \rightarrow E \rightarrow C \rightarrow B \rightarrow A \rightarrow D = 27 \text{ km}$$

$$S_5: D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D = 23 \text{ km}$$

$$S_6: D \rightarrow C \rightarrow B \rightarrow A \rightarrow E \rightarrow D = 29 \text{ km}$$

→ by comparing min is 23 km

$$\therefore S_5: D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$$

neighbours of S_5 have cost 22 km

$$\therefore \text{Move from } 23 - 22 = 1$$

$$\therefore \text{Final local search cost} = 22 \text{ km}$$

cost of every neighbouring solution ≥ 22

Final local optimum = 22 km